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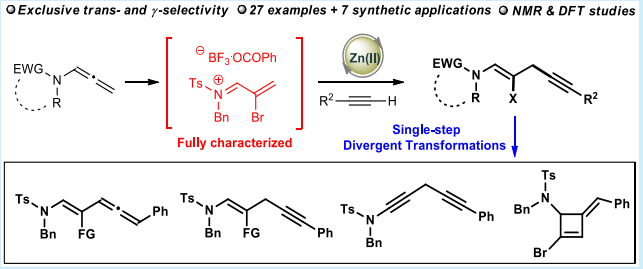
Characterization and Utilization of the Elusive α,β-Unsaturated *N*-Tosyliminium: the Synthesis of Highly Functionalizable Skipped Halo Enynes

Tapas R. Pradhan, Mukti Paudel, Jordan L. Harper, Paul Ha-Yeon Cheong,* and Jin Kyoon Park*

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ABSTRACT: A formal haloalkynylation of allenamides has been described for the synthesis of highly stereo- and regioselective skipped halo enynes. Exclusive γ-regioselectivity is achieved through the intermediacy of a conjugated *N*-tosyliminium intermediate—direct evidence for the formation of which was validated by NMR and HRMS. Quantum mechanical computations reveal that the reactive intermediate geometry is key to controlling the 1,2- or 1,4-regioselectivity of alkyne interception. Divergent access to elusive unsaturated systems has also been reported.



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N-Acyliminium ions (NAIs), which are more electron-deficient than the Mannich reagents, have been extensively employed in the construction of alkaloids, natural compounds, and drug-like molecules (Scheme 1A, right; synthetic routes to NAIs).[1] Recently, α,β-unsaturated NAIs have also been successfully utilized in many selective additions and Diels−Alder reactions for the synthesis of complex fused heterocycles.[2] Although the *N*-sulfonyliminium ions have been relatively less exploited,[3] the α,β-unsaturated analogue ions (i.e., of acyl and sulfonyl iminiums) by various electrophilic activations are proposed as potential intermediates to explain unique regio- and stereoselectivity in recent allenamide chemistry (Scheme 1A, left).[4] Despite vast applications of the iminium ions by trapping with nucleophiles, neither α,β-unsaturated *N*-acyliminium nor *N*-sulfonyliminium ions have been spectroscopically observed, presumably due to high reactivity and instability, to our knowledge.[4k] Notably, the first precedent in the detection of NAIs, generated from α-alkoxycarbamates using $BF_3{\cdot}OEt_2$, by NMR was reported in 1992 by Yamamoto et al.[5]

1,*n*-Enynes are important synthetic building blocks in modern organic synthesis with many elegant methods for their preparation, as well as great versatility, which allows for ready diversification into other functional groups.[6] Alkynylation of alkynes or allenes[7,8] is a well-established method for the generation of 1,*n*-enynes since the first discovery by Trost in 1990,[8a] and several catalytic variants have also been reported for the synthesis of 1,3- or 1,4-enynes.[9] A direct synthetic approach to fluorinated skipped enynes via hydroalkynylation of *gem*-difluorinated cyclopropanes has also recently been reported by Fu.[10] In the context of adding haloalkynes into unsaturated C−C bonds, Jiang reported Pd(II)-catalyzed haloalkynylation cross-coupling reactions of internal alkynes and cycloalkenes.[11,12]

We recently reported a ligand-controlled strategy for the regio- and stereodivergent synthesis of 1,3- or 1,4-enynamides from allenamides (Scheme 1B).[13] Continuing to pursue the selective synthesis of functionalizable 1,4-enynamides, we hypothesized that nucleophilic alkynes could trap the α,β-unsaturated *N*-tosyliminium ion known to be generated from allenamides and *N*-halosuccinimides;[4d−i] no fruitful results were witnessed however, although cases involving the intermediacy of such α,β-unsaturated *N*-tosyliminium ions are proposed in many selective addition reactions to allenamides such as haloimidation,[4d] haloamination,[4e] haloalkoxylation,[4f] and haloarylation.[4h] Careful screening for the creation of α,β-unsaturated *N*-tosyliminium ion reactive intermediates using various *N*,*O*-aminal adducts and Lewis acids reveals the first NMR and HRMS detection of **A**, which was generated from the reaction of **2a** and $BF_3{\cdot}OEt_2$, and followed by successful trapping with a catalytically generated alkynylzinc species (Scheme 1C).[14] The successful detection of **A** is attributed to the stable "*ate*" complex formation of nucleophilc benzoic acid with BF_3 (see Figure 2).

Our investigation commenced with the screening of various different *N*,*O*-aminals (**2**), obtained by reacting allenesulfonamide **1** with NBS and acid or alcohol,[15] with phenyl acetylene

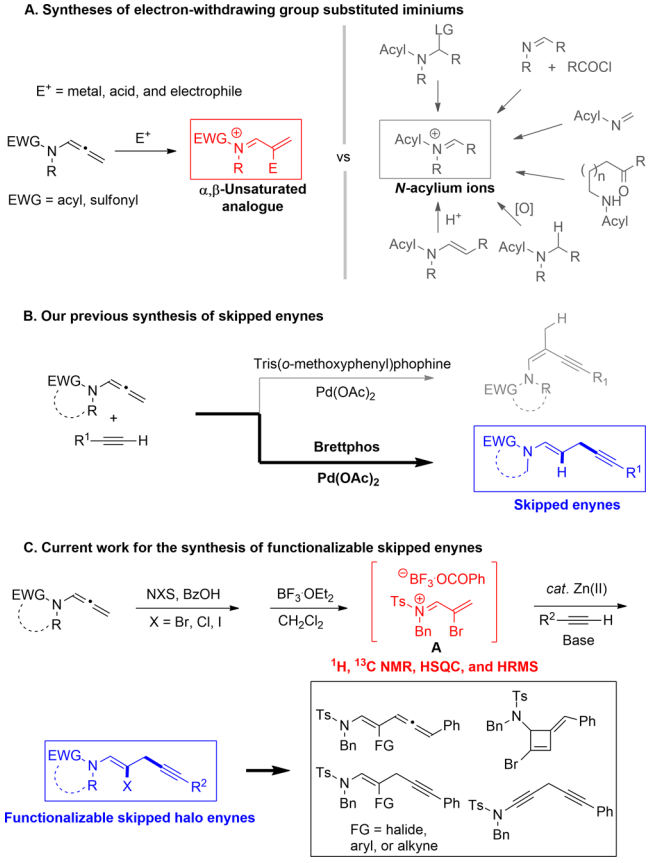
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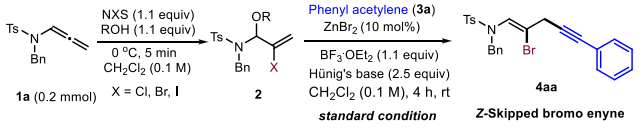
Scheme 1. (A) General Allenamide Activation and *N*-Acyliminium Formations, (B) Our Previous Alkynylations, and (C) Synthesis of (*Z*)-Halo Skipped Enynes



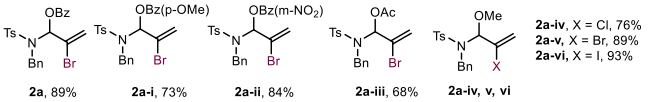
(**3a**) under Lewis acidic conditions using metal salt (Scheme 2).[16] After thorough screening, we found that the reaction of **2a** with $ZnBr_2$ (10 mol %), $BF_3{\cdot}OEt_2$ (1.1 equiv), and Hünig's base (2.5 equiv) gave the highest yield of **4aa** in 94% isolated yield with exclusive *Z*-selectivity (see Supporting Information for details of stereochemistry assignment). γ-Alkynylation product **4aa** using acetate-counterpart **2a-iii** was isolated in 73% yield, outcompeting the α-alkynylation product (entry 2). Further screening using other α-bromo-*N*-sulfonyl-*N*,*O*-aminals (**2a−i** and **2a-ii**) including α-halo-*N*,*O*-aminals (**2a-iv** to **2a-vi**) under identical conditions did not reveal a good overall yield of **4aa** in comparison to aminal (**2a**) having a benzoate leaving group (entries 1, 3, and 4). Control experiments showed that subtracting either reagent, including deviation of their equivalent amount, is not beneficial (entries 5−10). The attempt to employ Li-acetylide was futile, yielding an intractable mixture of compounds (entry 11). A sequential one-pot reaction gave a poor yield (33%) presumably due to the residual imide; however, its removal by basic workup doubled the overall yield of **4aa** (entry 12). As expected, the controlled experiments without Zn-salt or Hünig's base did not provide **4aa** (entry 13).

With reaction conditions optimized, we next investigated the scope of alkyne coupling partners **3** (Scheme 3) and are happy to report that a broad substrate scope for both aryl and aliphatic alkynes are tolerated. The reaction is also tolerant of a variety of substituents located in various positions on the aryl rings, affording the analogous halo enynamides (**4aa−4ai**) in up to 95% yield (Scheme 3A). Of note is that the reaction

Scheme 2. Search for Optimum Reaction Conditions[a]



A. Yield of the 1st step (Synthesis of various N,O-aminals)

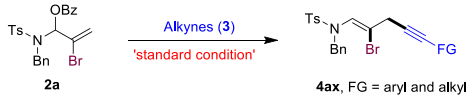


B. Optimization of the 2nd step (Synthesis of 4aa)

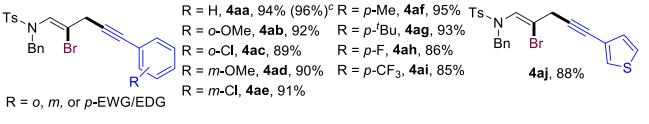
Entry	Variations from the "standard" condition	conversion (%)	Yield of **4aa** (%) (from isolated **2**)[b]
1	none (from **2a**)	100	96 [94][c]
2	N,O-aminal **2a-iii** was used instead	100	73
3	N,O-aminals (**2a-i** and **2a-ii**) were used instead	100	87 and 93, respectively
4	N,O-aminals (**2a-iv to 2a-vi**) were used instead	100	73, 81 and 67, respectively[d]
5	Zn-salts ($ZnCl_2$, ZnI_2 or $Zn(OTf)_2$) instead of $ZnBr_2$	5-47	up to 43 (maximum from ZnI_2)[e]
6	Cu-salts (Cu(I)Cl/Br/I/Tc or $Cu(OTf)_2$) instead of $ZnBr_2$	0-100	up to 81 (maximum from CuBr)[e]
7	$BF_3{\cdot}OEt_2$ replaced by TMSOTf, TfOH or *p*-TSA	0-62	up to 16 (maximum from TfOH)[e]
8	Hunig's base replaced by Et_3N, DBU, or *i*-Pr_2NH	32-45	up to 43 (maximum from DBU)[e]
9	other solvents were used instead of CH_2Cl_2	0-100	up to 67 (maximum from acetonitrile)[e]
10	lowering equivalent of $BF_3{\cdot}OEt_2$ from of 1.1 to 0.2	10-58	up to 54 (maximum from 0.75 equiv)[e]
11	Li-acetylide of **3a** was used instead of $ZnBr_2$	100	0 (unidentified mixture of compounds)
12	From **1a** (after basic work-up of **2a**)	100	67 [66][c]
13	Without $ZnBr_2$ or Hunig's base	100	0

[a]Unless otherwise mentioned, all reactions were carried out using 0.1 mmol of **2** and 0.13 mmol of **3a** under the conditions described in Scheme 1. [b]Yields are determined by NMR analysis using CH_2Br_2 as an internal standard. [c]Isolated yield. [d]Yields of **5a** (73%) and **5b** (67%) are shown together (see Scheme 4D for comparison). [e]For more extensive information, see Supporting Information.

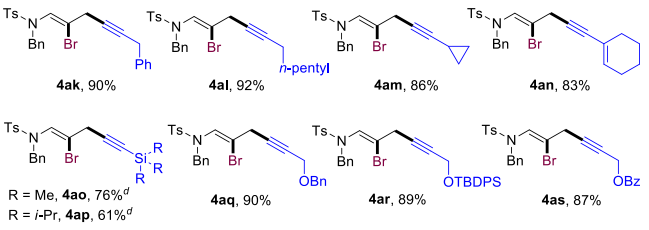
Scheme 3. Scope of Alkynes in Formal Bromoalkynylation of 1a[a,b]



A. Scope of aryl alkynes (10 nos. of bromo enynamides, up to 95% isolated yield)



B. Scope of aliphatic alkynes (9 nos. of bromo enynamides, up to 92% isolated yield)

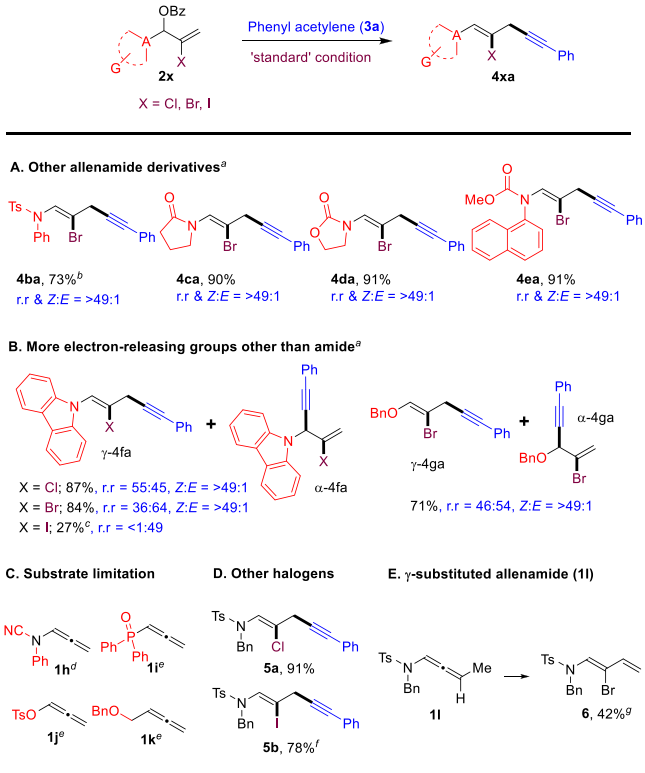


[a]All reactions were carried out using 0.2 mmol of isolated **2a** and 0.26 mmol of **3** under the standard condition. [b]Isolated yield of **4ax**. [c]Yield of **4aa** from 5 mmol scale. [d]Reaction was left for 12 h for complete consumption of **2a**.

tolerates halogenated alkynes (F and Cl), enabling the synthesis of halo enynes with easily modifiable reaction sites.

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Scheme 4. Effect of Allene Substitution and Halo Substituents in Formal Haloalkynylation with 3a[a]



[a]The r.r and *E*:*Z* ratio were determined from ^{1}H NMR (>49:1 means no detection of other isomer in NMR). [b]Alkynylation was performed without isolation of unstable bromobenzoyloxylation product, and the yield mentioned is the overall yield. [c]NMR yield of the α-alkynylation product as it could not be isolated from other unidentified compounds. [d]*N*,*O*-Aminal was successfully formed. [e]Alkynylation was unable to perform, as the bromobenzoyloxylation was failed. [f]Reaction was continued for 12 h for complete consumption of starting material. [g]Reaction performed after basic workup of *N*,*O*-aminal.

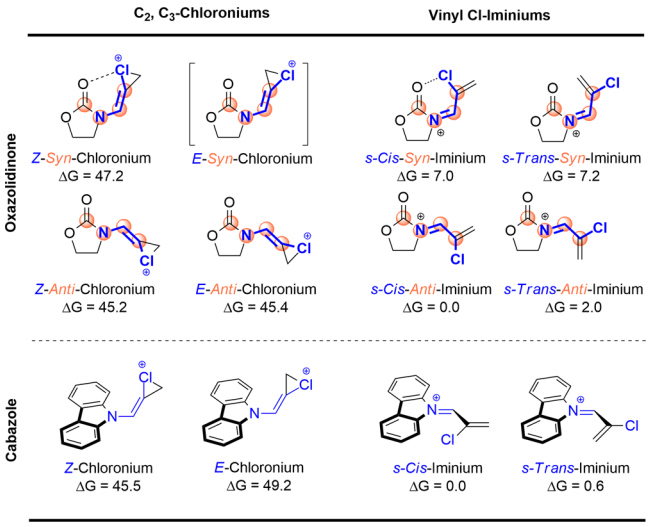


Figure 1. Calculated free energies of chlorinated reactive intermediates.

Bromoalkynylation to form **4aa** was efficiently scaled up to 5 mmol scale, and the product was obtained in 96% yield without chromatographic purification. Heteroaryl alkyne **3j** is

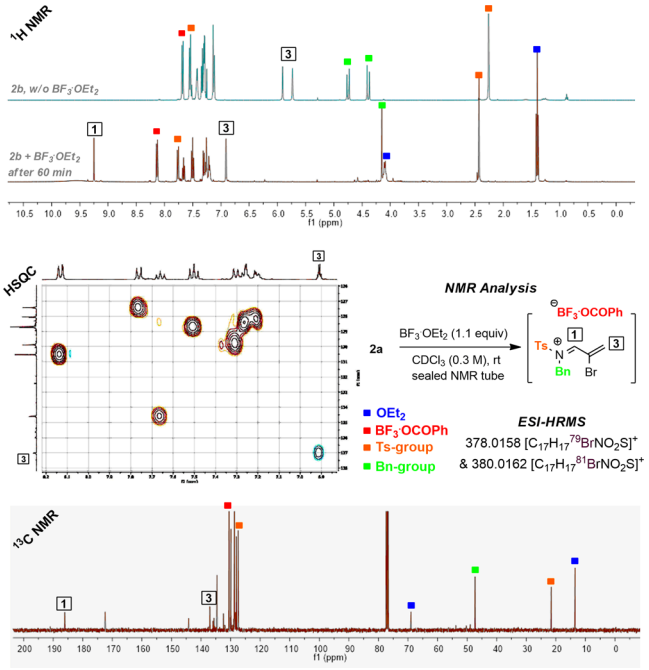


Figure 2. Characterization of *N*-tosyliminium intermediate **A** (NMR and HRMS).

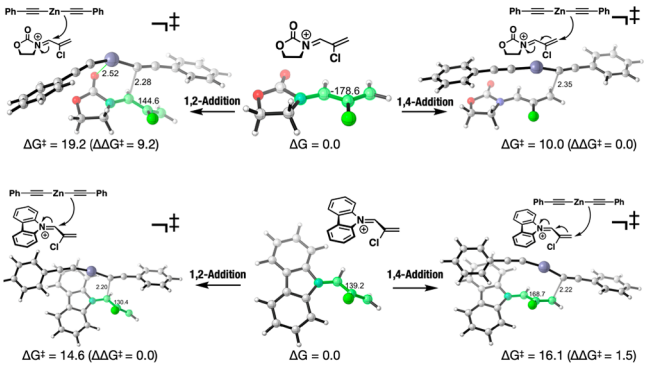


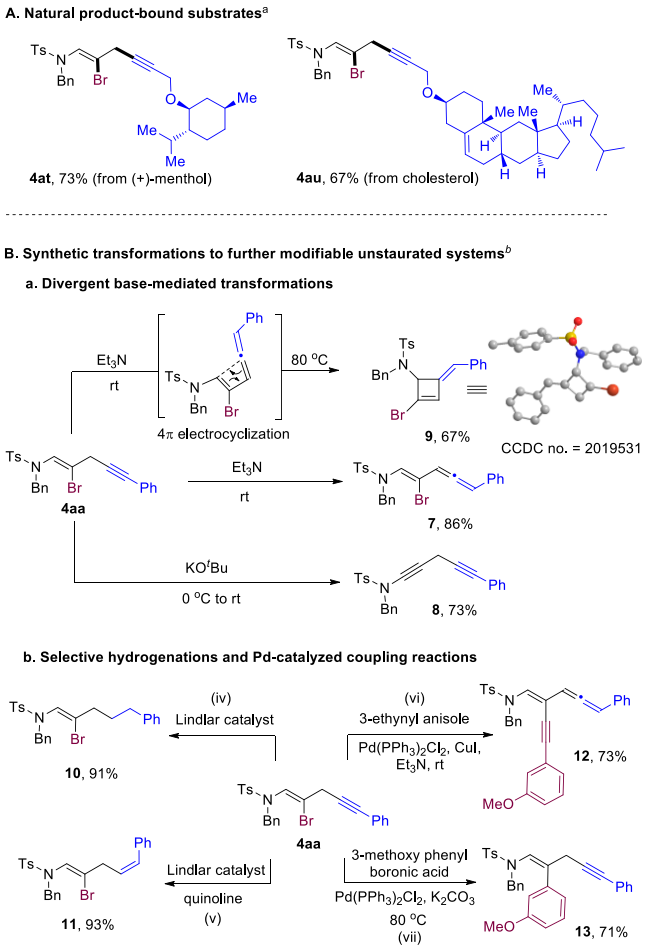
Figure 3. Reactive chlorination intermediates, and 1,2 and 1,4 addition transition states.

also a suitable substrate for this transformation, as are simple aliphatic alkynes (**3k**, **3l**), strained ring alkynes (**3m**), and 1,3-enyne (**3n**), which all smoothly gave the expected products in good yields (**4ak**−**n**). Silylated alkynes (**3o**, **3p**) also gave the desired products (**4ao**−**p**) in moderate yields, although a slightly extended time is required in comparison with other alkynes. This result is likely due to β-carbocationic stabilization by the proximal silyl functional group. Other acid- or base-sensitive functional groups including benzyl-, silyl-, and benzoyl-protected propargylic alcohols (**3q**−**3s**), were also well tolerated, highlighting the versatility of this reaction for modification of hydroxyl-based moieties.

The requirements of allenic substituents to have a group for effective electronic balance around the allenic skeleton were more rigid, and we noticed that it is essential for γ-selective alkynylation (Scheme 4). In addition to the tosyl group (**1a** and **1b**[17]), cyclic amide groups were also viable, giving the corresponding γ-alkynylation product (**4ca**) in good yield. Moreover, both cyclic and acyclic carbamate substituents did not affect the stereo- and regioselectivity of the product (**4da**, **4ea**), which further imitates the predictable electronic bias

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Scheme 5. Synthetic Applications[a,b]



[a]Haloalkynylation was performed under the optimized conditions. [b]All reactions were carried out using 0.1 mmol of **4aa** under the described conditions. Reagents and conditions: (i) Et_3N (1.5 equiv), THF, rt, 16 h, 86%. (ii) K^{t}OBu (1.1 equiv), THF, 0 °C to rt, 2 h, 73%. (iii) Condition (i) was employed at 80 °C for 24 h. (iv) Lindlar's catalyst (cal. 5 mg), H_2 balloon (1 atm), toluene, rt, 16 h, 91%. (v) Lindlar's catalyst (cal. 5 mg), Quinoline (20 mol%), H_2 balloon (1 atm), toluene (0.3 M), rt, 12 h, 93%. (vi) 3-Ethynyl anisole (1.3 equiv), $Pd(PPh_3)_2Cl_2$ (5 mol%), CuI (10 mol%), Et_3N/THF (1:2), rt, 12 h, 73%. (vii) 3-Methoxy phenyl boronic acid (1.5 equiv), $Pd(PPh_3)_2Cl_2$ (5 mol%), K_2CO_3 (3 equiv), 1,4-dioxane/H_2O (2:1), 80 °C, 24 h, 71%.

seen with sulfonyl substituents (Scheme 4A). To further probe the substituent effect controlling the γ-regioselectivity, we subjected *N*,*O*-aminals of allenamine **1f** (carbazole bearing bulkier N-substituent) and allenyl ether **1g** without any electron-withdrawing group (EWG), to our bromo-alkynylation conditions (Scheme 4B). Interestingly, mixtures of nonisolable regioisomeric products were obtained (with an α-alkynylation product as the major product for the Br-substituent and exclusive one for the I-substituent), signifying the requirement of a suitable electron biasing element on the nitrogen atom to achieve the γ-regioselectivity. While attempts to functionalize other allene derivatives such as those bearing a nitrile (**1h**), phosphine oxide (**1i**), *o*-tosyl (**1j**), and one carbon homologated allenyl ether (**1k**) failed (Scheme 4C). This process was nonetheless amenable to the use of other halogens (Cl, I) with chloro- and iodo-alkynylation products obtained in good yields, albeit slightly longer reaction times were required for **5b** indicating the halo participation is a reasonable proposition (Scheme 4D). A substituted allenamide **1l** led to a vinylogous elimination product **6**[17] (Scheme 4E).

To discover the origins of the observed regioselectivity, we turned to computations.[18] We first computed the ground state oxazolidinone and carbazole allenamide substrates and their halogenated reactive intermediates (Figure 1),[19] specifically the C1−C2 chloroniums, the C2−C3 chloroniums, and the vinyl chloride-iminiums. **C1−C2 Chloroniums** for both oxazolidinone- and carbazole-derived allenes spontaneously decomposed to vinyl chlorides, and the **C2−C3 Chloroniums** were energetically unreasonable (ΔG > 40 kcal/mol). In contrast, the vinyl chloride iminiums of them were found to be energetically accessible. These results suggest that the vinyl chloride iminiums are operative reactive intermediates in these reactions, not the chloronium ions. The possibility of halogen bonds was also probed for all these intermediates where geometrically possible, and in our hands, such interactions did not substantively improve the stabilities.

To acquire experimental validation of these computational analyses, NMR studies were then conducted for the benzoic acid derived *N*,*O*-aminal **2a** and methanol-derived *N*,*O*-aminal **2a−v** adducts (Figure 2). To our surprise, we were able to successfully elucidate the sulfonyl iminium structure at room temperature for the first time. In the case of **2a**, after treatment of 1.1 equiv of $BF_3{\cdot}OEt_2$ at room temperature, the appearance of a new peak at 9.25 ppm and the downfield shifting of both vinylic protons from 5.73 and 5.91 ppm to 6.91 ppm clearly indicate the formation of the iminium species. The analogous experiment involving **2a−v** resulted in a complex spectrum, revealing the highly reactive nature of the iminium derived from the corresponding *N*,*O*-aminal **2a−v**. This is presumably due to the more Lewis basic nature of the methoxide counterion compared to the benzoate counterion in the case of **2a**. The stability and the existence of the sulfonyl iminium intermediates were further corroborated through the appearance of *m*/*z* peaks in ESI-HRMS, corresponding to both isotopic molecular ions.

The regioselectivity of nucleophilic addition may be explained by the ground state conformations of the vinyl chloride iminiums. The vinyl chloride of the oxazolidinone iminium is planar at 179° and conjugated, while the vinyl chloride of the carbazole iminium is significantly bent out of planarity at >40° and out of conjugation. In the carbazoles, the steric repulsion between the vinyl chloride and the fused benzene prevents them from achieving planarity. These structures suggest therefore that the carbazole iminiums would most likely undergo 1,2 nucleophilic additions, while the oxazolidinone iminium would prefer 1,4 additions. Indeed, computed transition structures were in agreement with these observation and hypotheses. The carbazole iminium preferred to undergo 1,2-addition over the 1,4 by 1.5 kcal/mol, and in contrast, the oxazolidinone iminium preferred to undergo a 1,4-conjugate addition by 9.2 kcal/mol (Figure 3).[20] The computed selectivity values were in agreement with experiments, where the oxazolidinone additions gave exclusively 1,4 addition, whereas the carbazole additions gave a mixture. Moreover, the oxazolidinone has a significantly lower addition barrier by 4.6 kcal/mol compared to the carbazole, matching experiments that showed the additions of the former were rapid whereas the latter were sluggish.

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The synthetic applicability of this method can be easily exploited, illustrated, for example, in Scheme 5. In employing propargylic ethers as the alkyne coupling partner, bromo enynamides **4at** and **4au** were synthesized in acceptable yields, which showcases the late-stage diversification of natural products L-(+)-menthol and cholesterol (Scheme 5A). Further synthetic transformations can also be achieved, as shown in Scheme 5B: **4aa** was smoothly transformed into allene enamide (**7**) via organic base (Et_3N) mediated isomerization and skipped diynamide (**8**) through inorganic base (K^tOBu) mediated dehydrohalogenation (Scheme 5B-i and -ii). Interestingly, a one-pot base-induced isomerization and 4π-electrocyclization enabled synthesis of cyclobutene amide **9** (Scheme 5B-iii), the structure of which was confirmed by X-ray crystallography (CCDC number: 2019531). Furthermore, semi- and full-reduction of **4aa** using Lindlar's catalyst were also feasible to selectively hydrogenate the C≡C triple bond furnishing **10** and **11**, while keeping intact the enamide unsaturation (Scheme 5B-iv and -v). Notably, the C−Br bond of **4aa** could be modified to allene appended conjugated ynenamide (**12**) through a one-pot two-step transformation, and arylated-skipped ynenamide (**13**), by stereospecific Suzuki coupling (Scheme 5B-vi and vii).

In summary, an efficient formal haloalkynylation protocol has been reported. Studies on the electronic effects of N-substituents shows that the haloalkynylation process occurs through electronic cooperation with a neighboring halogen, thus enabling γ-selective interception of an alkynyl-nucleophile via generation of electrophilic vinyl halide species. More importantly, the intriguing functionalities of halo enynes pave the way to synthesize unsaturated synthons which can be further functionalized with ease.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c00103.

Experimental procedures, compound characterization data, and NMR spectra (PDF)

Accession Codes

CCDC 2019531 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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(18) All ground and transition state structures were computed at the ωB97XD/6-31G* level of theory with single-point energy refinements at the ωB97XD/6-311G*//SMD(CH_2Cl_2) level of theory.

(19) Z/*E* and s-*cis*/*trans* refer to the geometric relationship between the oxazolidinone carbonyl and the chlorine. *Syn*/*anti* refers to the orientation of the oxaxolidinone carbonyl and the alkene π bond. The C1, C2-chloronium structures are not stationary points on the potential energy surface and spontaneously rearrange.

(20) Alkynylzinc species are known to exist in solution in complex aggregation states. Efforts to compute the transition structures involving larger (dimer, trimer, and tetrameric) alkynylzinc aggregates were unsuccessful in our hands. Larger alkynylzinc aggregation states are expected to change the steric environment around the nucleophile and raise the barrier to additions, but not expected to change the selectivity, as the selectivity is largely dictated by the electrophile geometry and conjugation effects as described.

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